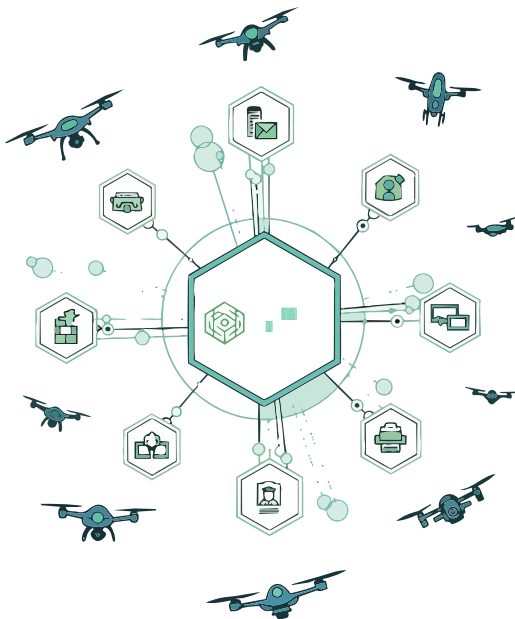


Building a Secure UAV Swarm Network Using Hyperledger Fabric

This hybrid security framework for UAV swarm communication integrates blockchain for trust management with cryptography for secure transmission.



Unmanned aerial vehicles (UAVs), commonly known as drones, are autonomous, remotely operated aerial systems capable of performing tasks without human intervention. A UAV swarm network consists of multiple drones that collaborate and communicate with one another to achieve a common objective, increasing their scalability, flexibility and efficiency in complex operations.

UAV swarms are today widely deployed in critical applications such as military surveillance, border security, and disaster management. These are also being used for rescue operations, environmental and habitat monitoring, and in agriculture.

However, the decentralised and dynamic nature of these UAVs makes them highly vulnerable to cyber-attacks such as Denial of Service (DoS), Sybil attacks and Man-in-the-Middle (MITM) attacks, which impacts their reliability, integrity and availability.

The multi-tier protection framework discussed below, which integrates Hyperledger Fabric blockchain and

cryptographic techniques such as AES-256 and ECC P-256, is an attempt at addressing these challenges. Blockchain ensures decentralised authentication and tamper-proof logging. Network anomalies are detected in real-time using the NSL-KDD dataset. Simulation results demonstrate an average blockchain latency of 45ms with less than 5% performance degradation, which makes this system suitable for real-time UAV swarm applications.

It is evident from Table 1 that Hyperledger Fabric provides better security than the public blockchain Ethereum. Hyperledger Fabric's Membership Service Provider (MSP) issues X.509 certificates to each UAV at registration time. So only registered drones can participate in the network. This simple feature helps to eliminate Sybil attacks at the network entry points; this is not possible in public blockchains like Bitcoin or Ethereum.

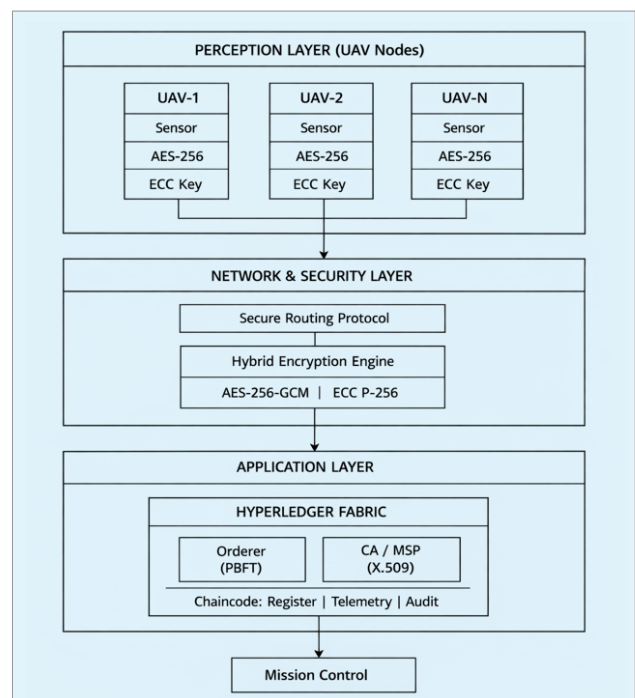


Figure 1: UAV swarm communication architecture